MERIDIAN ♥

Memory-Encoded Rituals for Intimacy across Distance in an Interactive, Asynchronous Neighbourhood

Jingyi Chen*
Independent Researcher
Shenzhen, China
jingyi@jingyifan.love

Yifan Yang†
Independent Researcher
College Station, Texas, USA
yifan@jingyifan.love



(a) The whole town, drawn in code (b) The square: guide, partner, cats (c) Prologue: the birthday cake (d) Finale: the wedding

Figure 1: Yi Town at a glance. (a) The town map: a 24×20 grid of 16-pixel tiles with nine numbered houses, one per level, painted entirely by code. (b) The fountain square in play, top after every level is cleared (gold flags, a cat sunbathing by the pond) and bottom halfway through; the player meets the partner, the guide Rayford and the cats. (c) The birthday prologue: nine candles in the dark at Shenzhen midnight, and ten strawberries set in a heart. (d) The wedding in the fountain garden, officiated by Rayford with the two cats as guests.

Abstract

How can shared memories become playable structures for partners who live in different time zones? We present MERIDIAN, a pre-deployment design case of Yi Town, a bilingual browser game developed by one member of a long-distance couple as a gift for the other. Nine levels translate personally meaningful events, places and cultural references into tasks within a navigable pixel town. A birthday prologue and a fictional wedding frame the intended experience. Drawing on research through design, we reconstruct four design changes from 109 development commits, 28 changelog versions and three design specifications. These records support six situated design commitments, including memory-derived mechanics, staged access, local progress storage and bilingual coverage. The implementation combines a union-based progress model with manually exchanged, checksummed codes and time-zone-dependent scenery. An implementation audit reproduced 153 passing unit tests and examined all 1,024 canonical progress codes, while identifying limits involving resets, unversioned level registries and residual transcription errors. The application has no author-operated backend for progress, but external media services and browser speech recognition limit its privacy and offline guarantees. No recipient evaluation or relationship outcomes are reported. The contribution is an inspectable account of design decisions and their trade-offs, including the tension between authoring a personal gift and preserving the recipient's freedom to interpret, decline or reshape it.

Keywords

intimate computing, long-distance relationships, autobiographical design, research through design, games for two, local-first software, bilingual interfaces, pixel art, ceremony, surprise

1 Introduction

Long-distance partners use communication technologies to maintain awareness, exchange affection and share ordinary activities [20, 25, 33]. Some report substantial intimacy despite geographical separation, but such findings do not imply that a particular technology improves a relationship [15, 31]. This paper asks a narrower design question: how can a couple's shared material become an asynchronous game, and what tensions arise when one partner authors both its narrative and its interaction constraints?

Yi Town is a browser game developed by Yifan for Jingyi and himself. They began their relationship on 20 September 2026 and live in Shenzhen, China, and College Station, Texas. Their clocks

*This manuscript reports the design of Yi Town before recipient evaluation.
†Corresponding author.

differed by thirteen hours during development; the difference becomes fourteen hours when Texas observes standard time. The game draws on shared songs, a Gomoku game, dated memories and a symbolic hand-drawn ticket representing a planned meeting. Players navigate a pixel town whose nine houses contain levels, while a birthday prologue, a letter and a fictional wedding organise the narrative (Figure 1). The game accommodates separate visits without requiring simultaneous participation; it does not prohibit the two people from playing at the same time.

We use Meridian as the name for this design case: Memory-Encoded Rituals for Intimacy across Distance in an Interactive, Asynchronous Neighbourhood. A *memory-encoded ritual* means a scripted action whose intended significance derives from shared autobiographical material. This is a design description, not evidence that the action becomes an established ritual in use. Likewise, *intimacy* names the intended qualities of recognition, affection and shared meaning; we do not operationalise or measure it as an outcome. The central design move is to make a memory shape an action or a rule, rather than only illustrate an interface.

The paper reports an artefact and a retrospective reconstruction of its development before the recipient's planned first play on 31 October 2026. It draws on research through design and autobiographical design, without claiming the extensive lived use associated with mature first-person studies [7, 26, 37]. Its contributions are:

(1) A documented design case that translates autobiographical material into nine game levels and staged ceremonies (Sections 3 and 4).
(2) Six situated design commitments and four traceable development episodes, with explicit distinctions between recorded changes, interpretations and unevaluated intentions (Table 2; Section 6).
(3) An implementation account and verification audit of local progress, manual code exchange and selected interface properties, together with their privacy and reliability limits (Section 5).

The application of established mechanisms such as union merge is an engineering choice within this case, not a new replication algorithm or a demonstrated intervention for relationships.

2 Related Work

2.1 Mediating Intimacy at a Distance

Vetere et al. studied how couples and families express intimacy and argued that technologies for strong ties should support the expressive, the physical and the playful, not only the informational [33]. Hassenzahl et al. reviewed 143 intimacy-mediating artefacts and grouped them into six strategies: awareness, expressivity, physicalness, gift giving, joint action and memories [11]. Many such systems are deliberately minimal channels, such as the Virtual Intimate Object, a single button whose meaning the couple negotiates [17]. Neustaedter and Greenberg found that some long-distance couples leave a video call running for hours to share "being together" rather than to talk [25]; CoupleVIBE conveyed a partner's location by vibration [3]; SyncDecor linked everyday appliances in two homes [32]. Yi Town sits in Hassenzahl's *memories* and *gift giving* strategies. The comparison concerns what the interface represents: a collection of shared references organised as tasks, rather than only a live awareness signal. This combination should not be read as evidence that autobiographical or bespoke intimate technologies are new.

2.2 Autobiographical and First-Person Design

Neustaedter and Sengers defined autobiographical design as design research drawing on genuine use by the people who built the system [26], and Desjardins and Ball documented its tensions, including the blurred line between designer and participant, and the intimacy of the data involved [5]. First-person methods include situated inquiry through the researcher's own experience; a small number of people alone does not establish methodological depth [6, 21]. Research through design treats the artefact itself as a carrier of knowledge [7, 37]. Yi Town uses autobiographical material and developer self-testing, but the present account precedes sustained use by the recipient. We therefore distinguish the making of the artefact from an evaluation of living with it.

2.3 Play, Ritual and Games for Two

Huizinga described play as a "magic circle" set apart from ordinary life [12]; Salen and Zimmerman made the circle a design tool [28]; Juul argued that games are half real rules and half fiction [16]. Sicart argued that play appropriates the world and its objects [30], and Isbister argued that games build connection between players through shared action [14]. Ludic design in HCI values curiosity and ambiguity over task efficiency [8, 9], and slow technology designs for reflection rather than throughput [10, 27]. Aron et al.'s procedure for generating closeness through escalating mutual disclosure [2] is, in a sense, a level structure. Yi Town takes the game form literally: locks, levels, a guide, a prologue and a finale. Bespoke ritual technologies already explore connection across distance [18]. Our case instead organises selected memories and imagined futures as a sequence of game actions. Whether that sequence becomes meaningful ritual requires evidence from use.

2.4 Local-First and Backend-Free Progress

Local-first software seeks to combine collaboration with user control and durable access to data [19]. State-based conflict-free replicated data types (CRDTs) converge when replicas combine compatible state through an appropriate merge; the grow-only set uses union [29]. Yi Town adopts this operation for level completion and exchanges state manually. It supports local progress, rather than implementing every local-first ideal. Static hosting, remote media and browser-dependent recognition remain part of its infrastructure. Keeping progress local also does not protect personal images that are distributed as static assets; visual content can expose location information [35].

3 Design Context and Method

3.1 The Two of Us

Table 1 lists the dates that the system encodes. The couple speak Mandarin to each other; Yifan works in English, Jingyi in Chinese and English. Both use phones as their main devices, and Jingyi's

Table 1: Key dates encoded in the system (`src/core/config.ts`).

Date	Moment
2026-09-20	We met; day 1 of every counter in the town
2026-09-25	Yifan's birthday (Libra)
2026-10-17	First video call; the hand-drawn ticket CYF925
2026-10-23	First Gomoku game, on a Zoom whiteboard
2026-10-24	First commit of Yi Town
2026-10-31	Jingyi's birthday (Scorpio); intended first recipient play
2027-02-05	Planned meeting; symbolic ticket CYF925; countdown target

network in mainland China blocks some services (YouTube, Google Fonts) that Yifan's does not. These circumstances motivated bilingual text, alternative media providers and a phone-oriented interface. They are design requirements, not verified guarantees of regional availability or cross-device compatibility.

3.2 Scope, Corpus and Reconstruction

The design corpus comprises 109 commits dated 24–28 October 2026, 28 changelog entries from version 0.1.0 to 1.17.0, and three specifications for the site, quest and town. The development cut-off is commit `55440ae`; the manuscript inspected for revision is commit `94ce914`. Yifan directed the design, supplied the autobiographical material and tested both player roles, with substantial implementation assistance from an AI coding assistant. Jingyi is the intended recipient. This account reports no interview, diary, recipient play session or relationship measure. The two intended players therefore do not constitute a two-participant evaluation sample.

For the present reconstruction, we inspect the documented changes underlying four episodes: replacing Pictionary, changing the ring's completion condition, correcting represented routines, and revising the entry gate (Section 6). Each episode links a source change to an interpretation of the design trade-off. The selection illustrates decisions discussed in this paper; it is not an exhaustive coding of all commits or an independently coded thematic analysis. Version history establishes what changed and when, but cannot establish the recipient's response or the designer's unrecorded motivation. Interpretations remain open to alternative readings.

3.3 Knowledge Claims and Positionality

Research through design can make an artefact and its rationale available for scrutiny without claiming a universal solution [7, 37]. First-person inquiry also requires attention to whose experience becomes evidence [5, 6]. Here, the designer's familiarity with the relationship informs the content but cannot substitute for the other partner's perspective. Technical checks establish selected implementation properties. The six commitments below are situated design choices reconstructed from the artefact and log; they are neither empirically validated principles nor measured determinants of intimacy.

Table 2: Design commitments and the mechanisms that realise them.

Commitment	Realised by
D1 Memory-derived mechanics	Memories, cultural references and imagined futures become tasks (Table 3)
D2 Earned access	Ordered level registry; router guard; door-only entry; cats guarding the next locks
D3 Local progress	`localStorage` per player; grow-only set; progress codes; PWA offline cache
D4 Bilingual coverage	Matching non-empty dictionary keys; no independent language-quality evaluation
D5 Drawn in code	Sprites and tiles as strings; banner and icons generated from them
D6 Delayed disclosure	Lexical test over eligible dictionary keys, with explicit exclusions

3.4 Six Design Commitments

We articulate six commitments from this reconstruction of the five-day development period. Table 2 maps each to the mechanisms that realise it.

D1 Memory-derived mechanics.
Selected memories and shared references shape tasks. The Gomoku level reworks a prior game; the ring represents an imagined future rather than a past event.

D2 Staged access.
Ordered levels and door entry organise the narrative. These controls shape normal play, but are not security barriers or evidence of meaningful effort.

D3 Local progress.
Completion state is stored on-device and shared through a code, without an author-operated progress backend. External services remain involved in other features.

D4 Bilingual coverage.
Both dictionaries contain non-empty values for the same keys. Structural tests do not establish semantic equivalence or accessibility in either language.

D5 Code-defined pixel art.
Town tiles and character sprites are generated in code. Photographs, audio, rendered rings and other assets have separate provenance and reuse conditions.

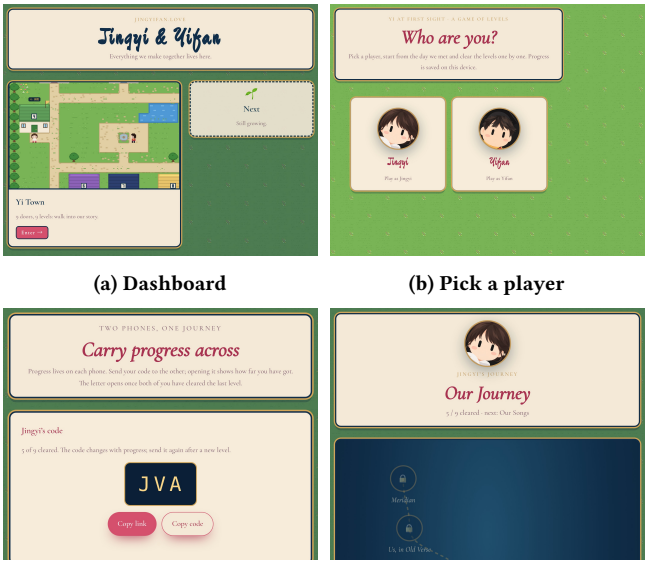
D6 Delayed narrative disclosure.
A lexical test checks selected dictionary entries for explicit references to the finale. It does not prove that every interface or asset preserves surprise.

4 The Experience

4.1 The Front Door

The site opens on a single question. The answer, a date both players know, is accepted in several spellings: its digits are compared against SHA-256 hashes [24]. Writing $\delta(a)$ for the digits of an answer a in order, H for SHA-256 and $\mathcal{A}$ for the four stored digests,

$$\mathrm{pass}(a) \iff H\big(\delta(a)\big) \in \mathcal{A}. \tag{1}$$



(a) Dashboard **(b) Pick a player**

(c) Progress code **(d) The journey list**

Figure 2: Around the town. (a) The dashboard lists every product under the domain. (b) Each device keeps one progress per player. (c) Progress crosses phones as a code (here JVA). (d) The journey as a list, for players who prefer it to the map.

Hashing changes the comparison representation, but the date also appears in the application's narrative and source material; it is not a secret protected by cryptography. Behind it is a dashboard that lists everything the couple build under the domain (Figure 2a), of which Yi Town is the first. Choosing the town asks *Who are you?* (Figure 2b). Progress is kept per player on each device, so both can play on one phone.

4.2 The Town

The town (Figure 1a) is a 24×20 grid of 16×16-pixel tiles. Paths connect nine houses around a fountain square and a pond. Each roof has its own colour and its level number, and a cleared house flies a gold flag. The player moves with the arrow keys, by tapping a tile (the avatar walks a shortest path to it), or with an on-screen d-pad. Walking into a door opens a talk box with the level's one-line story, for example "Our first game was on a Zoom whiteboard. Now you place the last stone."

Four kinds of character live in the town (Figure 3). The *partner* waits by the fountain and reports how far they have got ("I am waiting here. I have cleared 5 levels. You?"). *Rayford*, Yifan in a suit, is a scripted guide south of the fountain: he explains the project, lists the levels, hints at the next one and walks the player to its door. Two Ragdoll cats, *Coffee* (咖啡) and *Duoduo* (哆哆), guard the locks: Coffee sits at the next locked door and Duoduo at the one after. When a door opens they walk to their new posts, and with nothing left to guard they sunbathe by the pond. Bumping into a cat shows a photo of the real cat. Two signposts lead out of the square to the two cities (Section 4.4), and a red mailbox holds voice notes the partner has left.



Figure 3: Every character in Yi Town is a string of characters in the source. From left: Jingyi, Yifan, Rayford, Yifan as the baker, Jingyi crowned, the bride and the groom (four facings each), then the cats Coffee and Duoduo. Bodies are 12×12 pixels over four rows of shared legs.

The sky follows the clock in the player's own city: Jingyi walks the town in Shenzhen time and Yifan in College Station time, so at the same instant one sees noon and the other sees lit windows under a night wash. The whole site stands on the same pixel meadow, and every heading is a cream sign in the town's palette, so leaving a house never leaves the world.

4.3 Nine Levels

Table 3 lists the levels with their source memory, mechanic and input modality, and Figure 4 shows them in play.

1. The Day We Met. The first level shows the day count since 20 September, the time in both cities with day and night icons, and a countdown to the flight. To clear it, each player reads a vow, rewritten from the wedding formula as the vow of "becoming each other's partner once more", and says one line aloud: 我爱你，陈静仪，我愿意 ("I love you, Jingyi Chen, I do"), with the partner's full name. Browser speech recognition [34] hears Mandarin names inconsistently, so the matcher removes everything except Han characters and Latin letters and accepts any of eight or nine homophone spellings of each name (for example 陈静怡, 陈静宜). A typed fallback keeps the level playable in browsers that cannot listen.

2. Love Map. Six places matter to the couple: Texas, Los Angeles, Japan, Shenzhen, Hong Kong and New York. The street views are public photographs of well-known sites from Wikimedia Commons rather than the couple's own pictures, since a personal photograph can give away far more of a place than its owner intends [35]. Before each heart appears on a Leaflet map, the player names a street-level photograph from four options (Figure 4b), and a wrong pick reshuffles the options. Great-circle arcs connect the two home cities.

3. Tattoo. The player handwrites a line or a small drawing on an arm or a shoulder, drawn in SVG, and inks it. The tattoo is kept on the device and can be saved as an image to send.

4. Moments. The timeline holds eight dated memories, each with photos and a story in both languages, from hearts made from both sides of the ocean (day 21) to a heart drawn on the Gomoku board (day 35). The player orders five of them by tapping pairs to swap, then reassembles a 3×3 portrait of the partner.

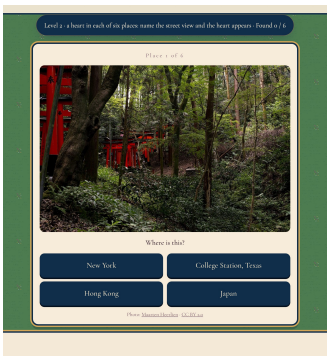
5. Gomoku. Each of ten positions on a 15×15 board has exactly one winning point for the player's own colour (purple for Jingyi, green for Yifan, as in their first game). The ten answers are the ten points of a heart; solved answers stay on the board in gold, so the heart draws itself as the player progresses. Positions are generated

Table 3: The nine levels: autobiographical material, shared references and imagined futures become tasks (D1). Modalities: *V* voice, *T* touch or pointer, *C* camera, *K* keyboard alternative available.

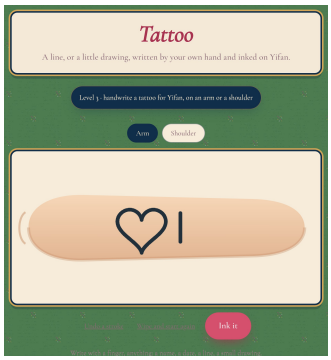
#	Level	Design material	Task (clear condition)	Input	Data
1	The Day We Met	20 September 2026; the vow	Answer a question, then say 我爱你，<name>，我愿意 aloud; homophones of the name accepted	V/K	1 vow per player
2	Love Map	Where each studied; where they will meet	Name six street views; each right answer drops a heart on the map to collect	T	6 places
3	Tattoo	A promise to write on each other	Handwrite a line on an arm or shoulder, then "ink it"	T	strokes
4	Moments	The first 40 days	Order five memories, then reassemble a 3×3 portrait of the partner	T	8 memories
5	Gomoku	First game on a Zoom whiteboard	Ten win-in-one endgames; the ten answers form a heart	T	10 positions
6	Our Songs	The six songs they share	Hear a passage, pick the title from four	T	6 songs
7	Libra and Scorpius	Their birth signs, neighbours in the sky	Join 21 stars along 20 edges	T/K	J2000 positions
8	Us, in Old Verse	Classical love poetry	Restore the missing half-line of five poems character by character	T	5 poems
9	Meridian	What comes next	Try the ring on through the camera and take a photo	C	3 GLB models



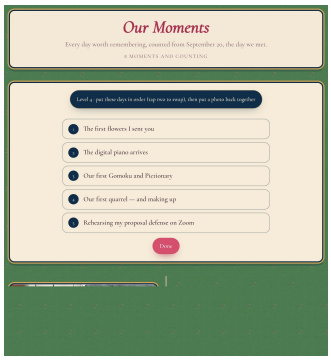
(a) 1 The Day We Met



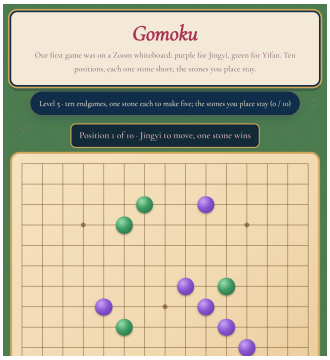
(b) 2 Love Map



(c) 3 Tattoo



(d) 4 Moments



(e) 5 Gomoku



(f) 6 Our Songs



(g) 7 Libra and Scorpius

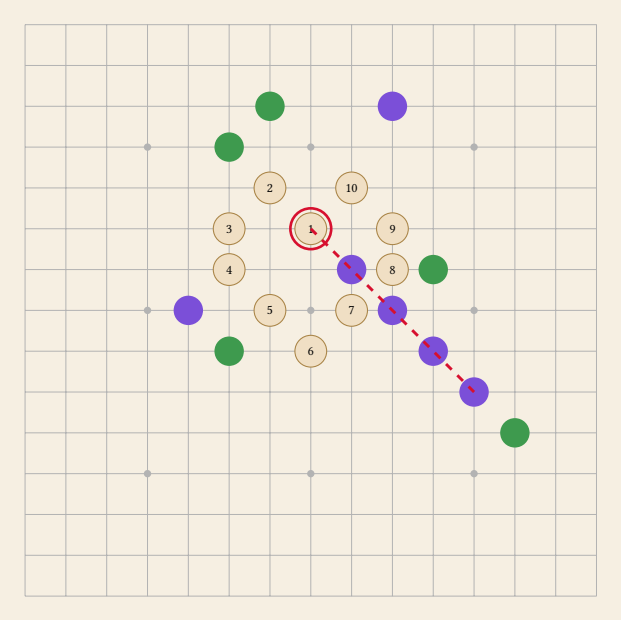


(h) 8 Us, in Old Verse

Figure 4: Eight of the nine levels in play (the ninth is shown in Figure 6). (a) Day counter, twin clocks and the countdown to CYF925. (b) Name the place in the street view (photograph: Maarten Heerlien, CC BY 2.0, via Wikimedia Commons). (c) A heart inked on an arm. (d) Put the first days in order (the photographs below are omitted here). (e) Position 1 of 10: one stone wins. (f) Name the song. (g) Libra's "claws" and Scorpius, before they are joined. (h) A line traditionally attributed to Su Wu, 结发为夫妻，恩爱两不疑 with its second half to restore.

from three patterns (a four closed at one end, a broken four, three-and-one) placed around each heart point in varied directions with decoy stones (Figure 5). Writing M for the mover's stones, $\mathcal{E}$ for the empty cells and $\mathrm{five}(\cdot)$ for "contains five in a row", a test requires of every position k that

$$\left|\{\, c \in \mathcal{E} : \mathrm{five}(M_k \cup \{c\}) \,\}\right| = 1, \qquad \{c_1^*, \ldots, c_{10}^*\} = \mathcal{H}, \tag{2}$$



15 × 15 board, rows 4–8 of the heart highlighted; purple: mover (Jingyi), green: partner

Figure 5: Ten answers draw a heart. The gold discs are the ten cells of $\mathcal{H}$ in the order they are found (Eq. 2); the stones are position 1 as generated in the source (`heart-1`): six purple stones for the mover, five green decoys, and the unique winning cell (red ring) completing a diagonal five. Later positions are built the same way around the other nine cells, so solved answers accumulate into the outline.

where c_k^* is that unique cell and $\mathcal{H}$ the ten cells of the heart. Gomoku is a first-player win under optimal play [1]; win-in-one puzzles avoid that question entirely and make every move feel like a find.

6. Our Songs. Six songs the couple share (among them 只对你有感觉 and 天下第一好) play a passage through the official embedded players: QQ Music for the player in China and YouTube for the player in the United States, each switchable. The site hosts no audio. For embedded players, the options unlock after eight seconds on screen because playback events are unavailable. This delay does not verify that a passage was heard.

7. Libra and Scorpius. Yifan is a Libra (25 September) and Jingyi a Scorpio (31 October); in the sky the two constellations are neighbours, and Libra's two brightest stars still bear the Arabic names of the Scorpion's claws, *Zubenelgenubi* and *Zubeneschamali*. The level projects 21 manually entered stars, six for Libra and fifteen for Scorpius, and asks the player to draw 20 edges. The source records coordinates as J2000 right ascension and declination; Meeus provides astronomical coordinate conventions [23], not provenance for these particular star entries. The chart is an illustration, not a computed view of either player's current night sky.

8. Us, in Old Verse. Five classical lines about love lose their second half, to be restored from a bank of characters: a farewell traditionally attributed to Su Wu, 结发为夫妻，恩爱两不疑; Guan Daosheng's *Song of You and Me*; a couplet of Li Shangyin; the paired birds of the sixth-century anthology *New Songs from a Jade Terrace* [4]; and an anonymous couplet. Each solved line shows a plain English gloss.

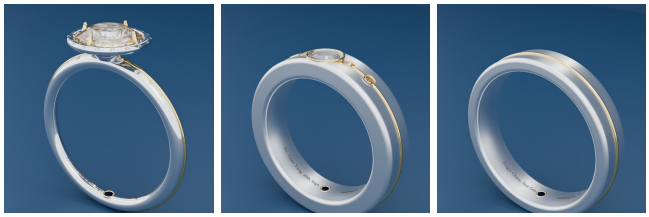


Figure 6: The Meridian pair, path-traced from the procedural Blender model: hers v1 (solitaire, halo and graduated shoulder stones), hers v2 (flush-set 4.2 mm stone in a 5 mm flat band), and his (brushed flat comfort-fit band). The gold meridian runs around each band.

9. Meridian. The last level is a pair of rings, called *Meridian*, modelled procedurally in Blender from a Python script (Figure 6). The pair share a signature: a yellow-gold line inlaid around a platinum band like a meridian on a globe, a hidden sapphire inside the bottom, and an inner engraving that carries the *other* person's name and birthday, the date they met and the coordinates of the airport where they will meet (40.64°N 73.78°W). Her ring exists in two designs, a solitaire on four compass prongs (v1) and the current flush-set stone sunk into his band's profile (v2); the page keeps both as a design history. The level is cleared only one way: the player tries the ring on through the camera and takes a photo (Section 5.5).

4.4 Two Days

Two signposts lead from the square to a day in each city (Figure 7). Each is a second map with the person's stops: Yifan's home, gym, his advisor's office, the golf course and the pool in College Station; Jingyi's home with the cats at the door, her drive, the office and Shenzhen Bay. Each stop has hours, and the person's sprite stands wherever the clock in their own city puts them. A partner who opens Yifan's day at 9:49 p.m. in Shenzhen finds him at 8:49 a.m. at his advisor's office. These are authored schedules, not live location tracking. They began as guesses and were corrected by the designer: Yifan drives rather than taking the bus; Jingyi drives a Tesla rather than taking the metro; and the Shenzhen Bay stop keeps a joke between them about where they will one day live.

4.5 Prologue and Finale

Two cutscenes frame the game (Figure 8). They run in a small pixel theatre: a 128 × 80 canvas scaled without smoothing, actors built from the town's sprite sheets, a darkness layer with candle-like glows, particles, a synthesised score, and a caption box for words and buttons.

The birthday prologue plays once per player before the town opens. In College Station it is 9:47 a.m.; in Shenzhen 10:47 p.m., 73 minutes before Jingyi's birthday. Yifan, in a chef's hat, bakes the cake against the Shenzhen clock. The player helps by whisking the batter (twelve taps), placing ten strawberries in the same heart that the Gomoku answers make, and watching nine candles lit (nine



(a) A day of Yifan's (b) A day of Jingyi's

Figure 7: The two cities as maps. The person stands at the stop whose hours contain the current local time in their own city: (a) Yifan at his advisor's office at 8:49 a.m. in College Station; (b) Jingyi at home at 9:49 p.m. in Shenzhen, the same instant.

levels; in Chinese 九 "nine" sounds like 久 "for a long time"). At the prologue's scripted Shenzhen midnight, rather than a real-time release gate, the lights go out, the door opens on the town ("In Yi Town there is no thirteen-hour time difference"), everyone sings, the player makes a wish and blows the candles out, by breathing into the microphone or holding a button.

The wedding opens after the last letter, once both players have cleared the last level. In the fountain garden at dusk, Rayford reveals why he had worn a suit all along: he is the master of ceremonies. The groom waits under the arch, the bride walks the aisle to Mendelssohn's *Wedding March*, the scripted sequence asks for both characters' "I do" responses, with speech recognition where supported, Coffee carries the rings down the aisle on a red pillow, the Meridian rings turn in 3D, and the scene ends in a kiss, bells, doves and a certificate to keep (Figure 9). Afterwards the town dresses the two sprites as bride and groom.

5 System

5.1 Architecture

Yi Town is a static single-page application in strict TypeScript, built with Vite and served from static hosting under the couple's domain (Figure 10). The deployment configuration targets GitHub Pages. There is no author-operated backend for progress. Each feature is a folder (`src/features/<name>/`) with its page module, styles, data, tests and a README, and each page is a function `render(root)` ⇒ `cleanup` loaded lazily by a path router, so the main bundle carries only the shell and the town. Heavy modules (Leaflet for the map, three.js for the ring, MediaPipe for the try-on) load only inside the level that needs them. A service worker precaches the shell and caches eligible media after retrieval. The cached town can run offline, but remote song embeds and browser-dependent speech recognition are not guaranteed offline; cache eviction and previously unfetched assets also limit availability.

5.2 The Quest Model

Let $L = (\ell_1, \ldots, \ell_9)$ be the ordered level registry and $C_p \subseteq L$ the set of levels player p has cleared on a device. A level is unlocked iff it is first or its predecessor is cleared:

$$\text{unlocked}_p(\ell_i) \iff i = 1 \ \vee \ \ell_{i-1} \in C_p. \tag{3}$$

Clearing adds to C_p and nothing ever removes from it, except an explicit reset. The next level to play is the first one not yet cleared, $\text{current}(p) = \ell_{\min\{i : \ell_i \notin C_p\}}$. The router enforces the lock, and a second guard requires that a level be entered through its door: walking onto the door sets a one-shot token τ in `sessionStorage`, and a level route without the token redirects to the town, so that

$$\text{enter}_p(\ell_i) \iff \text{unlocked}_p(\ell_i) \ \wedge \ \tau = \text{route}(\ell_i). \tag{4}$$

Figure 11 shows the whole sequence, including the two scenes gated on both players. Level numbers are derived from the registry, so inserting a level (the Poems level was inserted before the last one during development) renumbers every house, sign and string at once; a test fails until the new house has a door that the town's path search can reach.

5.3 Progress without a Server

Each phone holds its own C_p. Because C_p only grows, it is a grow-only set (G-Set), the simplest state-based CRDT [29]: the merge of two replicas is their union,

$$C_p \sqcup C'_p = C_p \cup C'_p, \tag{5}$$

which is commutative, associative and idempotent. Replicas converge if all updates are eventually exchanged, both phones use the same level registry and no reset occurs. Repeated imports are idempotent; an old code cannot remove completion flags, but incomplete exchange can leave the two views different. The only channel needed is one the couple already use every day: a chat message.

Progress codes. A code encodes one player and a bitmask of cleared levels, written in base 36 after a player letter $\lambda_p \in \{\texttt{J}, \texttt{Y}\}$ and followed by one check character χ (Figure 12):

$$m = \sum_{\ell_i \in C_p} 2^{\,i-1}, \qquad \text{code}(p, C_p) = \lambda_p \,\|\, \beta_{36}(m) \,\|\, \chi\big(\lambda_p \| \beta_{36}(m)\big), \tag{6}$$

where $\|$ is concatenation and β_{36} writes a number in base 36. The decoder accepts a string only if its last character equals χ of the rest, its first is a player letter and $0 \le m < 2^9$. With nine levels $m \le 511 = \text{E7}_{36}$, so a code is at most four characters: `JVA` says that Jingyi has cleared $m = 31$, the first five levels (Table 4). The check character follows the hybrid system ISO 7064 MOD 37,36 [13]. Writing $a_j \in \{0, \ldots, 35\}$ for the value of the j-th body character and starting from $p_0 = 36$,

$$s_j = \big((p_{j-1} + a_j) \bmod 36\big) \text{ or } 36 \text{ if } 0, \qquad p_j = 2s_j \bmod 37, \tag{7}$$

and the check character is the symbol of value $(37 - p_n) \bmod 36$. We checked the scheme exhaustively over all 1,024 valid codes. It rejects every one of the codes' single-character substitutions. Of the 2,943 transpositions of two different adjacent characters, it rejects all but five. One of those (`JI0`→`IJ0`) is still refused because `I` is not a player letter. The other four all swap a `0` and a `1` inside the mask



Figure 8: The two cutscenes, left to right. Top, the birthday prologue: the kitchen at 9:47 a.m. in College Station (10:47 p.m. in Shenzhen) with both clocks on the wall; whisking the batter; nine candles on the finished cake; singing at Shenzhen midnight with the cats. Bottom, the wedding: the guests take their seats; the bride walks the aisle; Coffee brings the rings; the Meridian rings in 3D over the ceremony.



Figure 9: The certificate at the end of the wedding, saved as an image through the phone's share sheet. It records the day of the ceremony, the day they met and the day they will meet.

(`J801`↔`J810`, `YE01`↔`YE10`): the rule above maps a running sum of 0 to 36, so a zero digit and its neighbour can trade places unseen. The exhaustive check covers 140,840 substitutions and 2,943 unequal adjacent swaps of canonical codes; the decoder rejects all substitutions and accepts four directed swaps (two bidirectional pairs). This is an error-detection check, not a user transcription study. An accepted typo may add unearned completion flags and unlock later content, including the finale. An earlier version used a plain weighted sum, which let some one-character typos through; it was replaced during a review pass, invalidating older codes. Codes are not authenticated: a person can construct completion for either player. They carry no schema version, so registry insertions or reordering can reinterpret bits. Reset is outside the grow-only model; importing an earlier code after reset restores its flags. A code may also travel in a link (`/sync?c=JVA`), exposing its payload

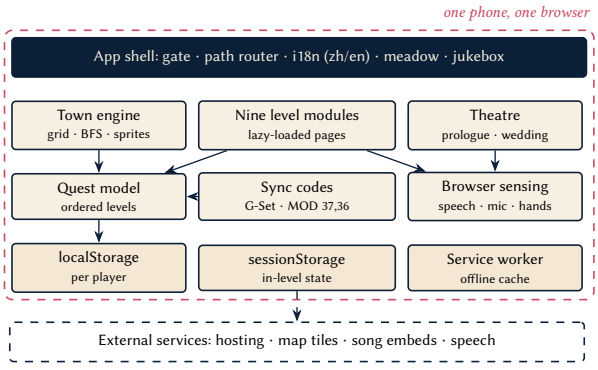


Figure 10: Architecture of application modules and local state. The dashed boundary marks browser components, not a confidentiality boundary: static assets are hosted remotely, external requests expose request metadata, and speech recognition may use a remote service. Progress crosses devices through a manually exchanged code.

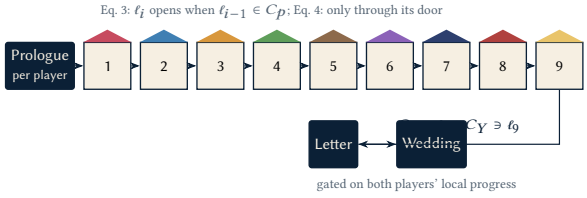


Figure 11: The sequence of play. Nine houses open in order per player; the letter and the wedding open only when both players' completion sets on the device contain the last level.

to the hosting request and potentially to browser history or the channel through which it is sent.

Table 4: Example progress codes: player letter, base-36 mask (Eq. 6) and the check character of Eq. 7.

Cleared	Mask m	Base 36	Code
Jingyi, levels 1–5	31	V	JVA
Jingyi, all nine	511	E7	JE7Z
Yifan, all nine	511	E7	YE7O
Yifan, none	0	0	Y0C

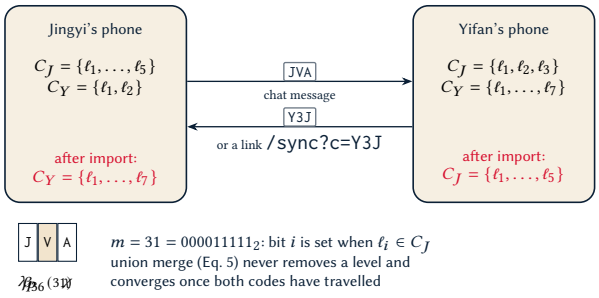


Figure 12: Progress without a server. Each phone keeps both players' completion sets; a short code (Eq. 6) carries one set across in a chat message or a link, and the receiver takes the union (Eq. 5). Below: the three fields of JVA.

Three layers of storage. Cleared levels live in `localStorage` under a key per player; half-finished state inside a level (the Gomoku position reached, the stars joined so far) lives in `sessionStorage`, so a reload or a language switch inside a level does not restart it, and it is dropped when the level is cleared; and a service worker keeps the site itself. When the browser blocks storage, an in-memory store keeps progress for the visit.

5.4 The Town Engine

The town is a grid of characters (`.` grass, `#` path, `~` water, `T` tree, `O` fountain, and so on). Movement is one tile per 190 ms with an ease-out interpolation $e(t) = 1 - (1-t)^2$ of the step's progress $t \in [0, 1]$; the pure functions `step` and `faced` decide movement and facing and are unit-tested, while a class wires them to a canvas and one animation loop. A tap on the map runs a breadth-first search over walkable tiles and queues the resulting steps. The camera follows the player. The same engine takes any map, so the two cities reuse it with their own grids, roads and a basketball court.

Art as text (D5). A sprite is an array of strings, one character per pixel, and a palette that maps characters to colours. A figure is a 12×12 body per facing over four rows of legs shared between facings, so a walk cycle is the same two leg frames under different bodies, and the side view is mirrored. Tiles (grass with flowers, paths that join their neighbours, roofs in each level's colour, lit windows) are painted procedurally. The README banner, the favicon and the app icons are generated from the same strings by scripts, so a new outfit appears everywhere at once.

Daylight by city. Let h_p be the current hour in player p's time zone (`Asia/Shanghai` or `America/Chicago`). The phase of the sky and

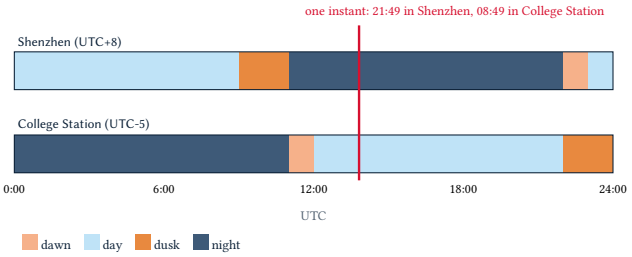


Figure 13: The two skies over one UTC day (Eq. 8), with the thirteen-hour offset in force during development. Night in one town overlaps day in the other for most of every day; the red line is the instant shown in Figure 7.

the person's place in a city day are

$$\phi(h) = \begin{cases} \text{dawn} & 6 \le h < 7, \\ \text{day} & 7 \le h < 17, \\ \text{dusk} & 17 \le h < 19, \\ \text{night} & \text{otherwise}, \end{cases} \tag{8}$$

$$\text{stop}(h) = \min \left\{ j : \exists\, [a, b) \in W_j,\ a \le h \bmod 24 < b \right\}, \tag{9}$$

where W_j is the set of hour windows of the j-th stop. Dawn, dusk and night each lay a coloured wash over the map (night: navy at 42% opacity) and light the windows. Because h_J and h_Y differ by the two cities' offset, the same instant falls in different phases for the two players (Figure 13).

5.5 Sensing in the Browser

Camera tracking and breath-amplitude processing run in application code on the device. Browser speech recognition has a different processing boundary.

Voice. The vow and the wedding's "I do" use the Web Speech API [34] in Mandarin. The implementation does not require on-device recognition; the API allows a remote recognition service unless local processing is explicitly required [34]. The application therefore cannot guarantee that voice samples remain on-device. A typed fallback is available for the vow. Recognition alternatives are joined into one string w and normalised by ν, which keeps only Han characters and Latin letters; with $\sqsubseteq$ for "is a substring of" and $\mathcal{N}_p$ the accepted spellings of the partner's name,

$$\begin{aligned} \text{vow}_p(w) \iff & \text{我爱你} \sqsubseteq \nu(w) \ \wedge \ \text{我愿意} \sqsubseteq \nu(w) \\ & \wedge \ \exists\, n \in \mathcal{N}_p : \ n \sqsubseteq \nu(w). \end{aligned} \tag{10}$$

Breath. Blowing out the candles reads the microphone with echo cancellation, noise suppression and gain control off. For each frame t of $N = 1{,}024$ samples $x_{t,i}$,

$$r_t = \sqrt{\tfrac{1}{N} \textstyle\sum_{i=1}^{N} x_{t,i}^2}, \qquad s_t = 0.6\, s_{t-1} + 0.4 \min(1, 5 r_t), \tag{11}$$

$$\text{blow}_t \iff s_t > 0.32.$$

We rely on a breath being a loud, broadband rush on a phone microphone; the threshold was set by hand and has not yet been calibrated across devices.

Hands. The try-on runs MediaPipe's hand landmarker [22, 36] on the camera stream, which returns 21 landmarks $\mathbf{k}_0, \ldots, \mathbf{k}_{20}$ per

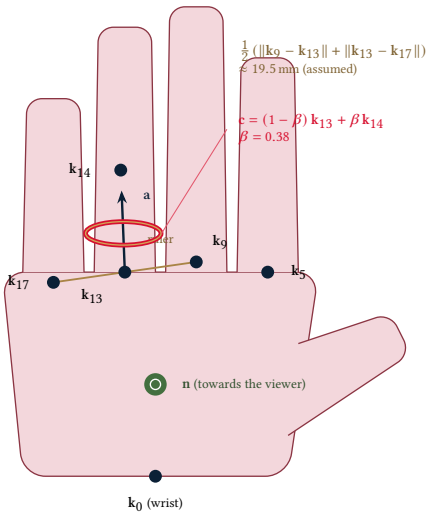


Figure 14: Placing the ring from hand landmarks (Eqs. 12–14). The band's axis follows $\mathbf{k}_{13} \to \mathbf{k}_{14}$, its centre sits 38% along that segment, the back of the hand is the palm normal $\mathbf{n}$ oriented by handedness, and the knuckle spacing $\mathbf{k}_9$–$\mathbf{k}_{13}$–$\mathbf{k}_{17}$ is the millimetre ruler.

hand (Figure 14). The ring's axis $\mathbf{a}$ follows the ring finger's proximal phalanx, its "up" $\mathbf{u}$ is the back of the hand, taken from the palm normal $\mathbf{n}$ with the sign given by the reported handedness, and its centre $\mathbf{c}$ sits a fraction $\beta = 0.38$ along the phalanx:

$$\mathbf{a} = \frac{\mathbf{k}_{14} - \mathbf{k}_{13}}{\|\mathbf{k}_{14} - \mathbf{k}_{13}\|}, \qquad \mathbf{n} = \pm \frac{(\mathbf{k}_5 - \mathbf{k}_0) \times (\mathbf{k}_{17} - \mathbf{k}_0)}{\|(\mathbf{k}_5 - \mathbf{k}_0) \times (\mathbf{k}_{17} - \mathbf{k}_0)\|}, \tag{12}$$

$$\mathbf{u} = \frac{\mathbf{n} - (\mathbf{n}\cdot\mathbf{a})\,\mathbf{a}}{\|\mathbf{n} - (\mathbf{n}\cdot\mathbf{a})\,\mathbf{a}\|}, \qquad \mathbf{c} = (1-\beta)\,\mathbf{k}_{13} + \beta\,\mathbf{k}_{14}, \tag{13}$$

and the rotation is the orthonormal basis $R = [\,\mathbf{u}\times\mathbf{a},\ \mathbf{u},\ \mathbf{a}\,]$. Scale comes from anatomy: the mean spacing of the middle, ring and little knuckles serves as the ruler, set to a constant of 19.5 mm (our working estimate for an adult hand, not a measured value),

$$\text{px/mm} = \tfrac{1}{2}\left(\|\mathbf{k}_9 - \mathbf{k}_{13}\| + \|\mathbf{k}_{13} - \mathbf{k}_{17}\|\right) / 19.5, \tag{14}$$

so nominal model dimensions map to the screen under this assumed scale. The display is a stylised overlay, not a validated measurement of ring size or fit. Position, rotation and scale are smoothed exponentially with $\alpha = 0.45$ per frame, $\mathbf{c}_t = \mathbf{c}_{t-1} + \alpha(\mathbf{c} - \mathbf{c}_{t-1})$ and $q_t = \text{slerp}(q_{t-1}, q, \alpha)$ for the rotation quaternion.

The shutter composes the photo with a caption, and a second card puts both players' try-on photos side by side.

5.6 Keeping a Secret with a Test (D6)

The ring is the surprise at the end, and an interface that hints at it spoils it. Rayford, the town, the progress page and the cats all speak before the ring is reached, and their lines were written and rewritten over days. Rather than rely on vigilance, a unit test scans every key in the bilingual dictionary, except the keys shown only after the ring, and fails if any contains 戒 (ring) in Chinese, *ring* as a word in English, or any word for wedding, marriage, proposal, bride or groom in either language. With K the set of dictionary keys, K_{after} those shown only after the ring, A the allow-list and $R_{\text{zh}}, R_{\text{en}}$ the two pattern tests, the test asserts

$$\forall k \in K \setminus (K_{\text{after}} \cup A): \quad \neg R_{\text{zh}}\big(D_{\text{zh}}[k]\big) \ \wedge\ \neg R_{\text{en}}\big(D_{\text{en}}[k]\big), \tag{15}$$

and a companion test asserts coverage, $\forall k \in K : D_{\text{zh}}[k] \neq \varepsilon \wedge D_{\text{en}}[k] \neq \varepsilon$ (D4). One element of A is listed by name: Jingyi's own joke on the Shenzhen Bay stop, which is about a flat, not a ring. The test establishes absence of specified lexical patterns in the selected dictionary entries. It does not inspect all assets, routes, source code or indirect visual cues, and therefore does not establish secrecy against inspection or unexpected navigation.

Table 5: Implementation summary (version 1.17.0, 28 October 2026).

Measure	Value
TypeScript source, excluding tests (lines)	8,414
CSS (lines)	1,080
Bilingual dictionary (lines)	791
Test files / unit tests	36 / 153
Commits / versions (24–28 Oct. 2026)	109 / 28
Main bundle (min. / gzip)	114 kB / 43 kB
Precached shell (service worker)	44 files, 302 KiB
Feature folders	24
Runtime dependencies	3 (Leaflet, three.js, MediaPipe)
Author-operated progress backend / app accounts	0 / 0

5.7 Implementation and Verification

Table 5 describes the version 1.17.0 implementation. A revision audit reran the existing suite: all 153 tests in 36 files passed, and the strict TypeScript check and production build succeeded. The main JavaScript chunk was 114.09 kB (42.58 kB gzip), and the service worker precached 44 entries totalling 302.12 KiB. Bundle sizes describe this build environment, not measured load time on users' phones. The large ring, map and try-on chunks are lazy loaded.

The audit also independently enumerated the progress-code cases reported above. Dictionary tests check matching non-empty keys; puzzle tests check properties such as unique winning moves. Passing these tests supports their asserted properties, not overall usability, privacy or an effect on intimacy. For example, the suite passes with 21 stars even though the earlier manuscript reported 22, because its star tests check consistency rather than that prose count.

The development records describe browser checks at desktop, 390-pixel and 320-pixel widths and a laptop-camera check of hand orientation. The inspected repository does not contain an archived per-release end-to-end report supporting the earlier claim that every release completed all nine levels at every stated width in both languages. That stronger claim is not retained. Systematic phone checks of speech, breath, camera completion and sharing remain outstanding, as does any recipient or relationship evaluation.

6 Reconstructing Design Decisions

Four documented episodes make the rationale inspectable. Commit identifiers distinguish recorded changes from our interpretations; neither a diff nor its message establishes a user outcome.

Replacing Pictionary (`7d2c659`*, 26 October).* The implementation removes a same-device drawing game and adds a song-identification level. Our interpretation is that separate recognition of a shared song better accommodates different visit times than taking turns on one screen. The diff establishes the replacement, not that the recipient preferred songs or that music increased closeness. A different reading is simply that the replacement reduced implementation complexity.

Requiring a try-on photo (`2bc3d07`*, 28 October).* The ring level's completion condition changes from viewing the reveal to capturing a try-on photo. This creates an artefact that can be shared, but also makes camera access and a capable device conditions for completing the narrative. The same mechanism can be read as participation or as an unwanted demand. No recipient account resolves that tension here.

Correcting represented routines (`1960404`*, 26 October).* The city-day descriptions change the transport and destination details to match the designer's account of the couple's routines. The record shows that plausible generated defaults required correction. It does not independently validate the representation of the other partner. Personal recognition is the design intention; verification by the represented person is still needed.

Revising the entry gate (`ccabd31`*, 28 October).* The review removes answer-revealing placeholder text and revises associated checks. This makes the recognition question less self-defeating, but does not turn a low-entropy, client-side date gate into access control. Its social role and its security properties must be evaluated separately.

Across these episodes, the transferable issue is how a technical rule expresses an assumption about the other person. The records support the existence of those assumptions and revisions; their acceptance, emotional meaning and durability require different evidence.

7 Discussion

7.1 From Representation to Action

The design links a shared reference to an action, then to a visible trace: a prior game becomes a win-in-one puzzle and its solution becomes one point of a heart. Compared with displaying the same memory as a caption, this adds an authored sequence of participation. The proposed design opportunity is to examine which part of a memory can become a rule without losing its meaning. The case demonstrates a concrete implementation of that move, not that it is more effective than a photo album, a message or another intimate artefact [11, 18]. The ring also exposes a boundary: an imagined future is not a remembered event, although both can organise narrative play.

7.2 Privacy Boundaries and Personal Material

The gate hashes a low-entropy date, whose value may also be inferred from the surrounding story. It deters casual entry but provides neither encryption nor dependable access control. Local completion flags avoid an author-operated progress database; they do not make hosted photographs, voice notes or letters private. The voice-note implementation, for example, references a static audio asset. Anyone authorised to obtain the deployed files or repository may obtain such material.

The browser boundary also differs by feature. Hand tracking and breath-amplitude calculation run locally, whereas speech recognition may contact a remote service. Tile and media requests expose request metadata to their providers. A local-first aspiration therefore needs a description of data flows, not only the absence of a custom backend [19]. An appropriate public research release would separate shareable code and synthetic examples from personal assets; intimate images can also reveal location context [35]. That separation is a release task, not a property established by the current artefact.

7.3 Asynchrony, Temporal Representation and Staleness

Each player's sky uses the clock in their configured city, while the partner avatar represents locally stored progress. The design supports visiting at different times, but an avatar's state may be old until the next manual exchange. Its city-day position follows an authored schedule, not sensed presence. This distinction matters: a representation intended as companionship can otherwise imply knowledge of the partner's current activity that the system does not have. The birthday scene's scripted clocks and the variable thirteen- or fourteen-hour offset are temporal design materials [10, 27], not evidence of temporal co-presence or synchronised participation.

7.4 Staged Access, Surprise and Freedom to Decline

Ordered access and delayed disclosure coordinate a surprise, but also let one partner determine what the other must do before reaching a letter or finale. Requiring a photograph illustrates the cost: people without camera access, or who prefer not to provide an image, cannot clear that level through the current completion rule. Similarly, a scripted "I do" is a game response, not evidence of consent to a relationship commitment. The present sequence does not model refusal as a separate successful outcome. A future revision should distinguish declining an action from failing a task and should offer an equivalent path that preserves narrative access.

The lexical test is useful as a writing regression check, but it cannot ensure surprise in a client-delivered game. Source, routes, media, screenshots and this paper disclose the finale. Any study of surprise must therefore record whether the recipient encountered these materials before play. Ambiguity may shape anticipation [8]; this case does not yet establish how it was experienced.

7.5 Bilingual Coverage and Situated Meaning

The same-key test prevents missing dictionary entries, but does not establish equivalent meaning, tone or task difficulty. The Chinese title 仪见钟情 substitutes a character from Jingyi's name into "love at first sight"; the English version explains the reference. Likewise, classical Chinese character-restoration puzzles retain different language demands even when their prompts have English versions.

A defensible claim is coverage of intended interface strings in two languages. Semantic quality and the accessibility of the resulting experience require a separate review, including by the intended recipient.

7.6 AI Assistance and Authorial Responsibility

The source and log document an AI-assisted implementation process in which the designer supplied requirements and corrected defaults. A correction shows a disagreement with generated output; it does not establish that all other generated details were accurate. The contribution is therefore the inspectable artefact and the rationale for selected choices, rather than a productivity claim inferred from five days of work. The corpus lacks an unassisted comparison and a complete record of human and model effort. Personal representation, verification and publication decisions remain the authors' responsibilities.

7.7 Limitations and Next Evidence

The principal limitation is evidential: developer playtesting is the only reported use, and no recipient experience, sustained practice or relationship outcome has been collected. The corpus is brief and selectively interpreted, with no independent analysis. Engineering checks are also bounded: tests do not establish deployment-wide privacy, all-browser compatibility or equivalent experience across languages; approximate hand scaling is not a fit measurement. Union merge depends on compatible registries and does not propagate resets. Remote media limits offline use and availability across regions.

A next stage could begin with a voluntary recipient walkthrough, recording what was understood, skipped, refused or reinterpreted, followed by a diary of revisits and code exchanges over time. These are proposed studies, not completed work. If participants are recruited or data are collected under an institution, the applicable ethics determination must precede that collection. A larger sample alone would not repair missing accounts of situated meaning, authorial power or failure.

8 Conclusion

MERIDIAN documents the pre-deployment design of a two-person game that turns autobiographical material into tasks, places and staged ceremonies. Its development record makes four design revisions inspectable, and its implementation audit establishes selected software properties. These support a situated account of how memories can shape playable rules. They do not establish that the game improves intimacy or that the recipient accepts its authored sequence. The central open question is how a personal gift can invite shared meaning while leaving the other person free to decline, reinterpret and reshape it.

Ethics and Publication Scope

This manuscript reports design and implementation, without recipient interviews or a relationship evaluation. Authorship, participation in play and permission to disclose personal material are separate decisions. Publication of identifiable narratives, dates, images, voices or represented routines requires the agreement of the people concerned; being a co-author is not a substitute for approving each category of disclosure. The manuscript does not report an institutional ethics approval or exemption. Any future study requires an appropriate determination before data collection.

Code and Data Availability

The inspected implementation is version 1.17.0 at commit `55440ae`; manuscript revision began from `94ce914`. The repository is located at https://github.com/rayford295/yi-town and was private when checked on 29 October 2026. A separately released, privacy-reviewed code archive and persistent archive identifier are not yet available. Personal photographs, voice notes and letters are not offered as a public research dataset. Repository visibility and consent must be rechecked when preparing any public release; the current statement does not promise unrestricted access to personal assets.

Declaration of Generative AI Use

The original manuscript reports use of Claude (Anthropic) through Claude Code for much of the software implementation, manuscript drafting and figure preparation. This revision additionally used Codex (OpenAI) for source inspection, editorial suggestions and implementation checks. Automated assistance and bibliographic metadata checks do not establish the truth of personal accounts or replace author review. The authors remain responsible for verifying the manuscript, attributions, reported evidence and permissions before submission. No AI system is listed as an author.

Acknowledgments

To Coffee and Duoduo, who guard the doors. The street-view photographs on the Love Map are from Wikimedia Commons and credited in the game; the *Wedding March* is a public-domain recording from the European Archive.

References

[1] Louis Victor Allis. 1994. *Searching for Solutions in Games and Artificial Intelligence.* Ph. D. Dissertation. Rijksuniversiteit Limburg, Maastricht, The Netherlands. doi:10.26481/dis.19940923la

[2] Arthur Aron, Edward Melinat, Elaine N. Aron, Robert Darrin Vallone, and Renee J. Bator. 1997. The Experimental Generation of Interpersonal Closeness: A Procedure and Some Preliminary Findings. *Personality and Social Psychology Bulletin* 23, 4 (1997), 363–377. doi:10.1177/0146167297234003

[3] Elizabeth Bales, Kevin A. Li, and William Griswold. 2011. CoupleVIBE: Mobile Implicit Communication to Improve Awareness for (Long-Distance) Couples. In *Proceedings of the ACM 2011 Conference on Computer Supported Cooperative Work (CSCW '11).* ACM, New York, NY, USA, 65–74. doi:10.1145/1958824.1958835

[4] Anne Birrell. 1982. *New Songs from a Jade Terrace: An Anthology of Early Chinese Love Poetry.* George Allen & Unwin, London.

[5] Audrey Desjardins and Aubree Ball. 2018. Revealing Tensions in Autobiographical Design in HCI. In *Proceedings of the 2018 Designing Interactive Systems Conference (DIS '18).* ACM, New York, NY, USA, 753–764. doi:10.1145/3196709.3196781

[6] Audrey Desjardins, Oscar Tomico, Andrés Lucero, Marta E. Cecchinato, and Carman Neustaedter. 2021. Introduction to the Special Issue on First-Person Methods in HCI. *ACM Transactions on Computer-Human Interaction* 28, 6, Article 37 (2021), 12 pages. doi:10.1145/3492342

[7] William Gaver. 2012. What Should We Expect from Research through Design?. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '12).* ACM, New York, NY, USA, 937–946. doi:10.1145/2207676.2208538

[8] William W. Gaver, Jacob Beaver, and Steve Benford. 2003. Ambiguity as a Resource for Design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '03).* ACM, New York, NY, USA, 233–240. doi:10.1145/642611.642653

[9] William W. Gaver, John Bowers, Andrew Boucher, Hans Gellerson, Sarah Pennington, Albrecht Schmidt, Anthony Steed, Nicholas Villars, and Brendan Walker.

2004. The Drift Table: Designing for Ludic Engagement. In *CHI '04 Extended Abstracts on Human Factors in Computing Systems*. ACM, New York, NY, USA, 885–900. doi:10.1145/985921.985947

[10] Lars Hallnäs and Johan Redström. 2001. Slow Technology – Designing for Reflection. *Personal and Ubiquitous Computing* 5, 3 (2001), 201–212. doi:10.1007/PL00000019

[11] Marc Hassenzahl, Stephanie Heidecker, Kai Eckoldt, Sarah Diefenbach, and Uwe Hillmann. 2012. All You Need is Love: Current Strategies of Mediating Intimate Relationships through Technology. *ACM Transactions on Computer-Human Interaction* 19, 4 (2012), 30:1–30:19. doi:10.1145/2395131.2395137

[12] Johan Huizinga. 1955. *Homo Ludens: A Study of the Play-Element in Culture*. Beacon Press, Boston. Originally published in Dutch, 1938.

[13] International Organization for Standardization. 2003. *ISO/IEC 7064:2003 Information Technology – Security Techniques – Check Character Systems*. Technical Report. ISO, Geneva.

[14] Katherine Isbister. 2016. *How Games Move Us: Emotion by Design*. MIT Press, Cambridge, MA.

[15] L. Crystal Jiang and Jeffrey T. Hancock. 2013. Absence Makes the Communication Grow Fonder: Geographic Separation, Interpersonal Media, and Intimacy in Dating Relationships. *Journal of Communication* 63, 3 (2013), 556–577. doi:10.1111/jcom.12029

[16] Jesper Juul. 2005. *Half-Real: Video Games between Real Rules and Fictional Worlds*. MIT Press, Cambridge, MA.

[17] Joseph 'Jofish' Kaye. 2006. I Just Clicked to Say I Love You: Rich Evaluations of Minimal Communication. In *CHI '06 Extended Abstracts on Human Factors in Computing Systems*. ACM, New York, NY, USA, 363–368. doi:10.1145/1125451.1125530

[18] David Kirk, David Chatting, Paulina Yurman, and Jo-Anne Bichard. 2016. Ritual Machines I & II: Making Technology at Home. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. ACM, New York, NY, USA, 2474–2486. doi:10.1145/2858036.2858424

[19] Martin Kleppmann, Adam Wiggins, Peter van Hardenberg, and Mark McGranaghan. 2019. Local-First Software: You Own Your Data, in Spite of the Cloud. In *Proceedings of the 2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward! 2019)*. ACM, New York, NY, USA, 154–178. doi:10.1145/3359591.3359737

[20] Danielle Lottridge, Nicolas Masson, and Wendy Mackay. 2009. Sharing Empty Moments: Design for Remote Couples. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '09)*. ACM, New York, NY, USA, 2329–2338. doi:10.1145/1518701.1519058

[21] Andrés Lucero, Audrey Desjardins, Carman Neustaedter, Kristina Höök, Marc Hassenzahl, and Marta E. Cecchinato. 2019. A Sample of One: First-Person Research Methods in HCI. In *Companion Publication of the 2019 Designing Interactive Systems Conference (DIS '19 Companion)*. ACM, New York, NY, USA, 385–388. doi:10.1145/3301019.3319996

[22] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Manfred Georg, and Matthias Grundmann. 2019. MediaPipe: A Framework for Building Perception Pipelines. arXiv:1906.08172

[23] Jean Meeus. 1998. *Astronomical Algorithms* (2nd ed.). Willmann-Bell, Richmond, VA.

[24] National Institute of Standards and Technology. 2015. *Secure Hash Standard (SHS)*. Federal Information Processing Standards Publication 180-4. NIST. doi:10.6028/NIST.FIPS.180-4

[25] Carman Neustaedter and Saul Greenberg. 2012. Intimacy in Long-Distance Relationships over Video Chat. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '12)*. ACM, New York, NY, USA, 753–762. doi:10.1145/2207676.2207785

[26] Carman Neustaedter and Phoebe Sengers. 2012. Autobiographical Design in HCI Research: Designing and Learning through Use-It-Yourself. In *Proceedings of the Designing Interactive Systems Conference (DIS '12)*. ACM, New York, NY, USA, 514–523. doi:10.1145/2317956.2318034

[27] William Odom, Richard Banks, Abigail Durrant, David Kirk, and James Pierce. 2012. Slow Technology: Critical Reflection and Future Directions. In *Proceedings of the Designing Interactive Systems Conference (DIS '12)*. ACM, New York, NY, USA, 816–817. doi:10.1145/2317956.2318088

[28] Katie Salen and Eric Zimmerman. 2003. *Rules of Play: Game Design Fundamentals*. MIT Press, Cambridge, MA.

[29] Marc Shapiro, Nuno Preguiça, Carlos Baquero, and Marek Zawirski. 2011. Conflict-Free Replicated Data Types. In *Stabilization, Safety, and Security of Distributed Systems (SSS 2011) (Lecture Notes in Computer Science, Vol. 6976)*. Springer, Berlin, Heidelberg, 386–400. doi:10.1007/978-3-642-24550-3_29

[30] Miguel Sicart. 2014. *Play Matters*. MIT Press, Cambridge, MA.

[31] Laura Stafford. 2005. *Maintaining Long-Distance and Cross-Residential Relationships*. Lawrence Erlbaum Associates, Mahwah, NJ.

[32] Hitomi Tsujita, Koji Tsukada, and Itiro Siio. 2008. SyncDecor: Communication Appliances for Virtual Cohabitation. In *Proceedings of the Working Conference on Advanced Visual Interfaces (AVI '08)*. ACM, New York, NY, USA, 449–453. doi:10.1145/1385569.1385656

[33] Frank Vetere, Martin R. Gibbs, Jesper Kjeldskov, Steve Howard, Florian 'Floyd' Mueller, Sonja Pedell, Karen Mecoles, and Marcus Bunyan. 2005. Mediating Intimacy: Designing Technologies to Support Strong-Tie Relationships. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '05)*. ACM, New York, NY, USA, 471–480. doi:10.1145/1054972.1055038

[34] WebAudio Community Group. 2026. Web Speech API. W3C Draft Community Group Report, 27 October 2026. https://webaudio.github.io/web-speech-api/ Accessed 29 October 2026.

[35] Yifan Yang, Siqin Wang, Daoyang Li, Shuju Sun, and Qingyang Wu. 2024. GeoLocator: A Location-Integrated Large Multimodal Model (LMM) for Inferring Geo-Privacy. *Applied Sciences* 14, 16 (2024), 7091. doi:10.3390/app14167091

[36] Fan Zhang, Valentin Bazarevsky, Andrey Vakunov, Andrei Tkachenka, George Sung, Chuo-Ling Chang, and Matthias Grundmann. 2020. MediaPipe Hands: On-device Real-time Hand Tracking. arXiv:2006.10214

[37] John Zimmerman, Jodi Forlizzi, and Shelley Evenson. 2007. Research through Design as a Method for Interaction Design Research in HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '07)*. ACM, New York, NY, USA, 493–502. doi:10.1145/1240624.1240704